Stochastic Signal Analysis

Chapter 1 Review Problem Set

Eight-paper edition – 30 problems – 10 days – 3 problems/day

This set is based on eight verified past papers (file codes 201812, 201912, 202104, 202112, 202212, 202312, 202412, 202512). Chapter 1 carries 27.500 primary marks per paper in this sample; the recent-four mean is 33.75. All current homework questions are included. Added older-paper questions fill topic gaps. Adaptations explicitly state extra assumptions and extensions. Complete one page at a time, leaving the blank space for your derivation.

2026-10-13

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Source tags distinguish verbatim homework, original exam excerpts, adaptations, and recall exercises. Day labels are a study schedule, not the order of the original exams.

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Problem 1 — Day 1

(Formula recall — §1.4.1 and §1.4.2)

Textbook sections: 1.4.1 / 1.4.2

For two random variables X and Y, write down the definition of each of the following quantities. Give both the general definition and the form used when a probability density function is available.

(a) the n-th moment about the origin, m_n;

(b) the n-th central moment, μ_n;

(c) the mean-square value and the root-mean-square value;

(d) the variance $D(X)$ and the standard deviation σ_X;

(e) the joint moment m_{kl} about the origin;

(f) the covariance $\mathrm{Cov}(X, Y)$;

(g) the correlation coefficient ρ_{XY}.

Finally, **derive** the identity

$$D(aX + bY) = a^2 D(X) + b^2 D(Y) + 2ab\,\mathrm{Cov}(X, Y).$$

Working space

Problem 2 — Day 1

(Textbook 1.21 / Required)

Textbook sections: 1.4.2

For the random variables X and Y, it is known that

$$EX = 1, \quad EY = 3, \quad D(X) = 4, \quad D(Y) = 16, \quad \rho_{XY} = 0.5,$$

Let $U = 3X + Y$ and $V = X - 2Y$. Find EU, EV, $D(U)$, $D(V)$, and $\mathrm{Cov}(U, V)$.

Working space

Problem 3 — Day 1

(2023–24 Q3(a)–(c); same density as 2018–19 Q2)

Textbook sections: 1.2.1 / 1.4.1 / 1.4.2

The joint probability density function of the random variables X and Y is

$$f_{XY}(x,y) = A\sin(x+y), \qquad 0 \le x \le \frac{\pi}{2}, \qquad 0 \le y \le \frac{\pi}{2}.$$

Find

(a) the coefficient A;

(b) the expectations $E(X)$ and $E(Y)$;

(c) the variances $D(X)$ and $D(Y)$.

Working space

Problem 4 — Day 2

(Final examination 2023–24, Q3(d))

Textbook sections: 1.4.2 / 1.4.3

For the same joint probability density function as in Question 3,

$$f_{XY}(x,y) = A\sin(x+y), \qquad 0 \le x \le \frac{\pi}{2}, \qquad 0 \le y \le \frac{\pi}{2},$$

find

(a) the joint moment $m_{XY} = E(XY)$;

(b) the correlation coefficient ρ_{XY}.

Hence state, with justification, whether X and Y are uncorrelated, whether they are orthogonal, and whether they are independent.

Working space

Problem 5 — Day 2

(Textbook Exercise 1.22)

Textbook sections: 1.4.3

The joint probability density function of the random variables X and Y is

$$f_{XY}(x,y) = \begin{cases} \dfrac{xy}{9}, & 0 \le x < 2, \quad 0 \le y < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Determine whether X and Y are **orthogonal**, whether they are **uncorrelated**, and whether they are **independent**. Justify each answer.

Working space

Problem 6 — Day 2

(Final examination 2024–25, Q1(b))

Textbook sections: 1.4.3 / 1.2.3

Two random signals are given by

$$X[n] = \sin(2\pi n + \phi), \qquad Y[n] = \cos(2\pi n + \phi),$$

where $\phi \sim U(0, \pi)$ and n is an integer. Calculate and discuss the **orthogonality**, the **uncorrelatedness** and the **statistical independence** of the two random signals. Justify your answer.

Hint: The deterministic identities for integer n simplify both signals a great deal before any averaging is done. Remember that orthogonality and uncorrelatedness are two different conditions, and that neither of them implies independence in general.

Working space

Problem 7 — Day 3

(Textbook 1.14 / Required)

Textbook sections: 1.2.1 / 1.4.2

The discrete random variables X and Y have the following joint probability mass function. Find:

$X\backslash Y$	-1	0	1
0	0.07	0.18	0.15
1	0.08	0.32	0.20

1. The joint distribution function and joint probability density function of X and Y;
2. The marginal probability mass functions of X and Y;
3. The probability mass function of $Z = XY$;
4. The correlation coefficient of X and Y.

Working space

Problem 8 — Day 3

(Final examination 2025–26, Q1(a)–(c))

Textbook sections: 1.2.1 / 1.4.2

A continuous-time random process is defined by throwing a coin once:

$$X(t) = \begin{cases} \sin(2\pi t), & \text{if the head is obtained,} \\ t^2, & \text{if the tail is obtained.} \end{cases}$$

The probabilities of obtaining a head and a tail are identical. Find

(a) the one-dimensional probability density function $f_X(x, 2)$;

(b) the two-dimensional probability density function $f_X(x_1, x_2; 1, 2)$;

(c) the covariance $C_X(t_1, t_2)$.

Hint: Because the process takes only two possible waveforms, every probability density here is a sum of Dirac impulses. Write the general expression in t_1 and t_2 first, and substitute the numbers only at the end.

Working space

Problem 9 — Day 3

(Final examination 2025–26, Q1(d))

Textbook sections: 1.2.1 / 1.2.3

A discrete-time random process is defined by independently throwing a coin repeatedly:

$$Y[n] = \begin{cases} \sin(2\pi n), & \text{if the head is obtained in the } n\text{-th throw,} \\ n^2, & \text{if the tail is obtained in the } n\text{-th throw.} \end{cases}$$

In every throw the probabilities of a head and a tail are identical. Find the two-dimensional probability density function $f_Y(y_1, y_2; 1, 2)$.

Working space

Problem 10 — Day 4

(Formula recall — §1.5.1)

Textbook sections: 1.5.1

Let X be a random variable with probability density function $f_X(x)$ and characteristic function $\phi_X(v)$.

(a) Write the definition of $\phi_X(v)$ as an expectation, and also as an integral.

(b) State the Fourier transform pair that relates $\phi_X(v)$ and $f_X(x)$, and state the uniqueness theorem.

(c) State Property 1: the characteristic function of a sum of independent random variables.

(d) State Property 2: the characteristic function of $Y = aX + b$.

(e) State Property 3: how the moments of X are obtained from the derivatives of $\phi_X(v)$ at the origin. Write the general formula for $E[X^k]$.

Working space

Problem 11 — Day 4

(Final examination 2022–23, Q1(b))

Textbook sections: 1.5.1

A random signal is given by

$$X(t) = U + t, \qquad t \in [-2, 2],$$

where U is a random variable uniformly distributed on $[0, \pi]$. Find the characteristic function of $X(t)$.

Hint: Find the characteristic function of U first, then use the property of Question 10(d). The result should depend on both v and t.

Working space

Problem 12 — Day 4

(Textbook 1.29 / Optional)

Textbook sections: 1.5.1

For each of the following characteristic functions of a random variable X, find its probability density function, mean, mean-square value, and variance.

1. $\phi(v) = 0.2 + 0.3\mathrm{e}^{\mathrm{j}2v} + 0.2\mathrm{e}^{\mathrm{j}4v} + 0.2\mathrm{e}^{-\mathrm{j}2v} + 0.1\mathrm{e}^{-\mathrm{j}4v}$;
2. $\phi(v) = 0.3\mathrm{e}^{\mathrm{j}v} + 0.7\mathrm{e}^{-\mathrm{j}v}$;
3. $\phi(v) = \dfrac{4}{4 - \mathrm{j}v}$;
4. $\phi(v) = \dfrac{\sin 5v}{5v}$.

Working space

Problem 13 — Day 5

(Formula recall · §1.6)

Textbook sections: 1.6

From memory, reproduce the following table for six typical distributions. For each one give the probability density function (or the probability mass function), the characteristic function, the mean and the variance.

	Distribution	Density / mass function	Characteristic function	Mean and variance
(a)	Uniform $U(a, b)$			
(b)	Exponential, parameter λ			
(c)	Gaussian $N(\mu, \sigma^2)$			
(d)	0–1 distribution, parameter p			
(e)	Binomial $B(n, p)$			
(f)	Poisson $P(\lambda)$			

Working space

Problem 14 — Day 5

(Textbook Exercise 1.17)

Textbook sections: 1.3.1

The output Y of a half-wave rectifier is related to its input X by the model

$$Y = g(X) = \begin{cases} X, & X \geq 0, \\ 0, & X < 0. \end{cases}$$

The input random variable X has probability density function $f_X(x)$ and distribution function $F_X(x)$. Find the probability density function $f_Y(y)$ of the output.

Hint: g is not one-to-one, and the whole negative half-line is mapped to the single value 0. Handle the continuous part and the discrete value separately, and expect a Dirac impulse.

Working space

Problem 15 — Day 5

(Adapted from 2023–24 Q2(c))

Textbook sections: 1.6 / 1.4.2

A zero-mean white noise $W(t)$ with PSD $N_0/2$ passes through $h(t) = e^{-t}u(t)$ to produce $X(t)$. Let $Y(t) = X(t) - X(t-T)$, with $T > 0$ and $N_0 > 0$. You may use $R_X(\tau) = N_0 e^{-|\tau|}/4$.

1. Do the stated assumptions imply that $Y(t)$ is Gaussian?
2. Under the additional assumption that W is Gaussian white noise, find the one-dimensional PDF of $Y(t)$.
3. Describe the case $T = 0$.

The extra Gaussian assumption in (2) is explicit; it was not stated in the original exam stem.

Working space

Problem 16 — Day 6

(Textbook 1.16 / Required)

Textbook sections: 1.3.2

Let $X \sim N(0,1)$ and $Y \sim N(0,1)$ be independent random variables. Define $U = X + Y$ and $V = X - Y$.

1. Find the joint probability density function $f_{UV}(u,v)$ of (U,V).
2. Determine whether U and V are independent.

Working space

Problem 17 — Day 6

(Formula recall · §1.2.1 and §1.6)

Textbook sections: 1.2.1 / 1.6

(a) Write the general bivariate Gaussian joint probability density function $f_{XY}(x, y)$ for means μ_X, μ_Y, variances σ_X^2, σ_Y^2 and correlation coefficient ρ.

(b) State the condition on ρ under which X and Y are independent, and explain why the condition works.

(c) Give the special case $\sigma_X = \sigma_Y = \sigma$ and $\rho = 0$.

(d) For a Gaussian random process $X(t)$, explain how the two-dimensional density $f_X(x_1, x_2; t_1, t_2)$ is obtained. State exactly which two quantities of the process determine it completely.

Working space

Problem 18 — Day 6

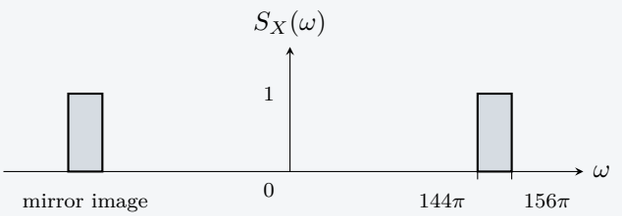
(Final examination 2025–26, Q4(b)–(c))

Textbook sections: 1.6 / 1.4.2

A zero-mean stationary Gaussian signal is given by

$$X(t) = i(t)\cos\omega_0 t - q(t)\sin\omega_0 t, \qquad \omega_0 = 148\pi,$$

where $i(t)$ and $q(t)$ are the in-phase and quadrature components. Its power spectral density $S_X(\omega)$ equals 1 for $144\pi \le |\omega| \le 156\pi$ and equals 0 elsewhere.



Find

(a) the joint probability density function $f_{iq}(i, q; t, t)$;

(b) the joint probability density function $f_{iq}\big(i, q; t, t + \frac{1}{12}\big)$.

Working space

Problem 19 — Day 7

(Textbook 1.6 / Required)

Textbook sections: 1.1.1

There are four batches of components. The first batch contains 2 000 components, of which 5% are defective. The second batch contains 500 components, of which 40% are defective. The third and fourth batches each contain 1 000 components, with approximately 10% defective. A batch is selected at random, and then a component is drawn at random from that batch. Find:

1. The probability that the selected component is defective.
2. The probability that the component came from the second batch, given that it is defective.

Working space

Problem 20 — Day 7

(Textbook 1.24 / Required)

Textbook sections: 1.4.4

The random variable X is uniformly distributed on $[0, a]$. Given X, the random variable Y is uniformly distributed on $[X, a]$. Find:

1. $E(Y \mid X)$, $0 \leq X \leq a$;
2. EY.

Working space

Problem 21 — Day 7

(Textbook 1.33 / Required)

Textbook sections: 1.5.2

The joint characteristic function of the random variables X and Y is

$$\phi_{XY}(u,v) = \frac{6}{6 - 2\mathrm{j}u - 3\mathrm{j}v - uv}.$$

Find:

1. The characteristic function of X;
2. The expectation and variance of Y.

Working space

Problem 22 — Day 8

(2018–19 Q1 (201812))

Textbook sections: 1.3.1 / 1.6

Let $X \sim U(0,1)$ and $Y = 5X + 1$. Find the PDF of Y, including its support. Check normalization.

Working space

Problem 23 — Day 8

(Adapted from 2019–20 Q1 (201912))

Textbook sections: 1.3.1 / 1.5.1 / 1.6

For this adapted question, explicitly assume $X \sim N(0,1)$ and $Y = 6X + 2$.

1. Find $f_Y(y)$ and $\phi_Y(v)$.
2. Explain why the original wording "Gaussian distribution in [0,1]" is ambiguous. If instead X is a standard normal conditioned on $0 \leq X \leq 1$, give the transformed PDF and a normalized integral for its CF.

Working space

Problem 24 — Day 8

(2019–20 Q2 (201912))

Textbook sections: 1.2.1

The joint PDF is $f_{XY}(x,y) = xe^{-x(y+1)}u(x)u(y)$. Find both marginal PDFs, verify normalization, and decide whether X and Y are independent.

Working space

Problem 25 — Day 9

(Adapted from 2020–21 Q4(b,c) (202104))

Textbook sections: 1.6 / 2.4.3

A zero-mean stationary Gaussian input has variance $\sigma_X^2 > 0$ and autocorrelation $R_X(\tau) = \sigma_X^2 e^{-a|\tau|}$. It passes through $h(t) = e^{-bt}u(t)$, where $a, b > 0$ and $a \neq b$. You may use the Chapter 5 result

$$R_Y(\tau) = \frac{\sigma_X^2}{b^2 - a^2}\left(e^{-a|\tau|} - \frac{a}{b}e^{-b|\tau|}\right).$$

Find the one-dimensional PDF of $Y(t)$ and $P\{Y(t) \geq 0\}$. Explain which assumption supplies Gaussianity.

Working space

Problem 26 — Day 9

(2021–22 Q1(c) (202112))

Textbook sections: 1.5.2 / 1.4.2

Let $X(t) = A + B\cos t$, with independent $A, B \sim N(0,1)$. For $Z = (X(0), X(\pi/2))^T$, determine the covariance matrix and the joint characteristic function $\phi_Z(u,v)$. Are the components independent?

Working space

Problem 27 — Day 9

(Textbook 1.9 / Optional)

Textbook sections: 1.1.2

The random variable X has the following probability mass function:

X	1	2	3
P	0.2	0.5	0.3

Find the probability density function and distribution function of X, and sketch their graphs.

Working space

Problem 28 — Day 10

(Textbook 1.11 / Optional)

Textbook sections: 1.1.2

The random variable X has probability density function $f(x) = ae^{-|x|}$. Find:

1. The coefficient a;
2. Its distribution function.

Working space

Problem 29 — Day 10

(Textbook 1.28 / Optional)

Textbook sections: 1.5.1

The random variables X_1, X_2, X_3 are mutually independent, with characteristic functions $\phi_1(v)$, $\phi_2(v)$, and $\phi_3(v)$, respectively. Find the characteristic function of each of the following random variables:

1. $X = X_1 + X_2$;
2. $X = X_1 + X_2 + X_3$;
3. $X = X_1 + 2X_2 + 3X_3$;
4. $X = 2X_1 + X_2 + 4X_3 + 10$.

Working space

Problem 30 — Day 10

(Textbook 1.30 / Optional)

Textbook sections: 1.5.1

Use the Fourier transform to derive the characteristic function of the uniform distribution.

Working space